

The University of Lahore, CS & IT Department

Object Oriented Programming

Lab Assignment # 01

Part -1: Hotel Management System

Start Date: 17/04/2025

Total Marks: 20

Due Date: 30/04/2025

Program: BSCS

Instructions

1. Understanding the problems is part of the assignment. So, no query, please.
 2. You will get zero marks if found any type of plagiarism or AI generated code.
 3. No submission after due date.
 4. Submit the .cpp file containing your code.
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Q1. Booking History & Rollback System using Stack

1. Objective

You are tasked with developing the Booking History Module of the Galaxy Hotel Management System using the Stack data structure. This module helps maintain the history of bookings and supports rollback functionality in case of booking cancellation or audit operations.

2. Scenario: Galaxy Hotel – The Booking Log Manager

The Galaxy Hotel is a luxury residence that maintains digital records of every booking for auditing and tracking purposes. To ensure that bookings can be reviewed, undone, or validated easily, they've hired you to build a booking history tracker using stacks.

Each successful booking made through the front desk or website should be pushed onto a stack. If a booking needs to be canceled or rolled back, the system should pop the most recent booking.

3. Core Requirements

Implement a Stack -Based Booking History System that stores the following details for each booking:

- Customer Name
- Room Number
- Room Type (Single, Double, Suite)
- Number of Nights
- Booking Date
- Status (e.g., Confirmed, Rolled Back)

4. Required Functionalities

4.1. Push Booking (Record New Booking)

- Add a new booking record to the stack.
- Display success message with all booking details.

4.2. Pop Booking (Cancel Most Recent Booking)

- Remove the most recent booking (i.e., the top of the stack).
- Display the removed booking and update its status as “Rolled Back”.

4.3. Peek Booking

- View the most recent booking without removing it.
- Useful for audit reviews.

4.4. Display All Booking History

- Traverse the stack from top to bottom and show all bookings with their current status.

4.5. Check If Stack is Empty

- Print appropriate message if no bookings are present.

4.6. Count how many bookings have been made for a specific room type (Single/Double/Suite).

4.7. Allow filtering bookings by customer name.

5. Program Requirements

- Use your own custom Stack implementation using arrays or linked lists (do not use STL or library stacks).
- The program should be menu-driven and user-friendly.
- Follow proper object-oriented structure using classes.